Joanna (Chiao Ting) Chang

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Healthcare Data Scientist with experience in medical and pharmacy claims, hospital utilization, and cost modeling across payer and provider settings. Experienced in working with large-scale medical and claims datasets to evaluate healthcare costs, risk drivers, and population health outcomes. Skilled at translating complex analyses into client-ready insights to support data-driven decision-making.

EDUCATION

Carnegie Mellon University, Heinz College - Dean's List — May 2026
Master of Science, Health Care Analytics & Information Technology — GPA: 3.82

National Cheng Kung University, College of Management - Phi Tau Phi Scholastic Honor, Ranked 1st — Jun 2016
Master of Business Administration, Business Administration and International Business — GPA: 3.98

SKILLS & CERTIFICATIONS

Programming & Tools: Python (Pandas, Scikit-learn, PyTorch, TensorFlow), SQL, R, Tableau, Power BI
Data Science & AI: Machine Learning (XGBoost, NLP, RAG/LLM), MLOps, ETL/ELT, Feature Engineering, Predictive Modeling, Inferential Statistics, Hypothesis Testing, SHAP, FAISS
Healthcare domain: Claims data (ICD-10, CPT, HCPCS, DRG, NDC), EHR/EMR, HIPAA
Cloud: AWS (**Certified Solutions Architect – Associate**), GCP, Docker, Kubernetes, GitHub Actions, CI/CD

PROFESSIONAL EXPERIENCE

Highmark Health — Pittsburgh, PA, USA
Data Science Team Lead (Capstone Project) — Jan 2026 - present
Modeling Longitudinal Cancer Cost Trajectories

- Built ETL pipelines in Python to integrate claims and clinical datasets for analytics and reporting
- Applied Group-Based Trajectory Modeling (GBTM) and time-series forecasting to identify longitudinal cost patterns
- Built XGBoost-based risk prediction models to classify near-term cost escalation risk
- Identified key clinical and non-clinical cost drivers through PCA and feature attribution

Carnegie Mellon University, AI Measurement Science & Engineering (AIMSEC) — Pittsburgh, PA, USA
Research Assistant (Data Science) — Jun 2025 - Aug 2025
Hospital-at-Home AI & Optimization | Applied ML Project

- Modeled Hospital-at-Home as a prediction-to-optimization pipeline, linking patient risk modeling with mixed-integer optimization for operational decisions
- Trained transformer/LLM models (structured EHR sequences) on 154K+ admissions extracted from MIMIC-IV via BigQuery to forecast care demand and utilization trends
- Evaluated system-level cost and service tradeoffs via simulation, across multiple acute conditions and distributed supply-chain configurations

Merck KGaA — Taipei, Taiwan
Senior Sales Specialist (Commercial Analytics) — Jan 2022 - Jun 2024

- Analyzed 180+ patient records annually to support clinical decisions with data-driven insights using Excel
- Drove a 32% increase in oncology product sales by leading a strategic, data-informed initiative
- Collaborated cross-functionally with APAC stakeholders to integrate analytics into medical education and planning

Johnson & Johnson, Medtech — Taipei, Taiwan
Sales Executive (Data-Driven Territory Strategy) — Nov 2019 - Dec 2021

- Leveraged surgical trend data to drive $400K in incremental product adoption
- Increased flagship product revenue by 6.7% ($4.3M) by translating analytical insights into actionable plans

PROJECT EXPERIENCE

Optimizing Diabetes Care: Predict-Then-Optimize Framework for Reducing Readmissions — Sep 2025 – Oct 2025

- Performed **ETL processes** on Encounter-Level diabetic patients' re-admission dataset from UC Irvine ML repository, developed an ML pipeline and conducted model evaluation
- Built **a predict-then-optimize ML framework** to reduce diabetic patient readmissions, achieving $2.2M savings with ROC AUC = 0.697
- Conducted **scenario modeling** for cost-effective interventions under budget constraints

EquiQuote: End-to-End ML Platform for Healthcare Analytics Mar 2025 – May 2025

- Designed and implemented a clinical **decision support system (DSS)** to predict insurance premiums, benchmarking Random Forest, tree-based, linear, KNN, and boosting models using R^2, RMSE, and MAE
- Optimized a Gradient Boosting Regressor in **scikit-learn** using **log transforms, hyperparameter tuning, and 5-fold cross-validation**, achieving $R^2 = 0.855$
- **Integrated explainable AI (SHAP)** for model explainability and deployed a **Streamlit** portal for real-time inference

FitScience Coach: RAG-Based LLM Learning Portal for Fitness & Nutrition Sep 2025 – Oct 2025

- Built an end-to-end **RAG pipeline using GPT-4o-mini, FAISS, and Hugging Face**; achieved 0.857 accuracy in RAGAs evaluation
- Developed an interactive dashboard using **Streamlit** for user queries, TDEE/BMR calculation, and personalized recommendations

COURSEWORK

Analytics & Modeling: Intermediate Statistics, Applied Econometrics I, Optimization, Decision & Risk Modeling Machine Learning with Python, NLP & LLM, Machine Learning in Production - AI Engineering, Generative AI
Technology & Data: Python Programming II, Database Management, NoSQL Database Management, Application of Natural Language Processing, Data Visualization with Python, Unstructured Data Analytics, Cloud Infrastructure and Services
Healthcare & Management: Health Systems, Health Policy, Healthcare Information Systems, Health Economics

LEADERSHIP & ACTIVITIES

Vice President, Badminton Club, Johnson & Johnson 2021
Taiwan Representative, 2019 APAC Sales Conference, Johnson & Johnson Vision Care 2018 - 2019
Vice Director, Student Association, NCKU Master of Business 2015 - 2016